

*MA.6.GR.2.2***Benchmark**

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

Benchmark Clarifications:

Clarification 1: Problem types include finding the area of composite shapes and determining missing dimensions.

Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a rectangle and triangle.

Clarification 3: Dimensions are limited to positive rational numbers.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.2, MA.6.NSO.2.3
- MA.6.AR.1.1
- MA.7.GR.1.2

Terms from the K-12 Glossary

- Algorithm
- Area
- Rectangle
- Quadrilateral
- Triangle
- Composite Figure

Vertical Alignment**Previous Benchmarks**

- MA.5.GR.2.1

Next Benchmarks

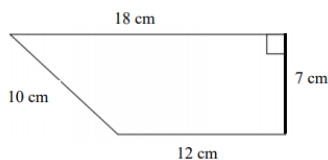
- MA.912.GR.4.4

Purpose and Instructional Strategies

In previous courses, students found the perimeter and area of rectangles with fractional or decimal sides. In grade 6 accelerated, students will apply the area of rectangles to find the areas of quadrilaterals and composite figures by decomposing composite figures into triangles and rectangles (*MTR.1.1*, *MTR.2.1*). Students will also extend this knowledge to decompose composite figures into triangles and quadrilaterals to find the area. In future courses, students will solve mathematical and real-world problems involving the area of two-dimensional figures.

- Instruction includes finding missing dimensions with quadrilaterals and composite figures (*MTR.1.1*, *MTR.5.1*).
- Instruction includes representing measurements for area as square units, units squared or units².
- When using rational numbers, instruction is restricted to numbers within the same form.
 - For example, if students are working with fractions, the side lengths will not include decimals. If students are working with decimals, the side lengths will not include fractions. Students should not be penalized though if they convert from one form to another when performing operations.

- Students should look for opportunities to either decompose or compose shapes to enhance their geometric reasoning.
 - For example, in the diagram below, students can solve by decomposing or by composing.



$$\text{Area} = (18 \cdot 7) - \frac{1}{2}(6 \cdot 7) = 105 \text{ cm}^2$$

$$\text{Area} = (12 \cdot 7) + \frac{1}{2}(6 \cdot 7) = 105 \text{ cm}^2$$

- Problem types include having students measure lengths using a ruler to determine the area.
- Instruction includes problems where multiple decompositions are possible so students understand the various pathways to a solution (MTR.5.1). If students decompose composite figures using quadrilaterals other than rectangles, this directly connects to benchmark MA.7.GR.1.2.

Common Misconceptions or Errors

- Students may invert the perimeter and area formulas.
- Students may incorrectly label all sides of the figure.
- Students may incorrectly identify a side of a triangle as the height.
- Students may overlook an unlabeled side.
- Students may incorrectly decompose a figure into irregular shapes.

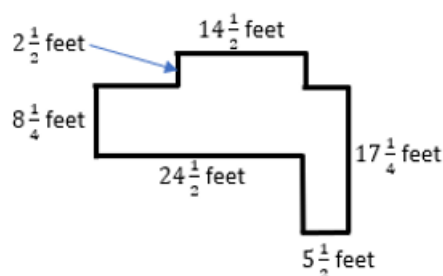
Strategies to Support Tiered Instruction

- Teacher relates the definitions of surface area and volume and co-creates an anchor chart to display in the room explaining each. Providing flashcards or cue cards with the formulas will help students in place of anchor charts when they are outside the classroom area.
- Teacher models the use of different color markers or pencils to match similar sides when decomposing figures. This will help students accurately label the sides of each shape.
- Use manipulatives or grid paper that students can measure to better understand there is a difference between a side length and the height in non-right triangles.
- Teacher models the use of manipulatives and geometric software to review the concept of perimeter or area.
- Teacher models the use of manipulative shapes to reinforce the sides of the pieces that make up a decomposed figure.

Instructional Tasks

Instructional Task 1 (MTR.1.1, MTR.2.1, MTR.5.1)

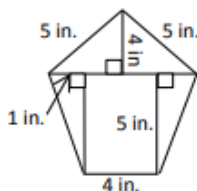
The diagram shows the dimensions, in feet, of the local playground. While playing with friends, Shona lost the key to her diary somewhere in the dirt. By composing or decomposing into rectangles, determine the maximum number of square feet that Shona may need to search to find the missing key? If she only searched one rectangular area, what is the least number of square feet Shona searched?



Instructional Items

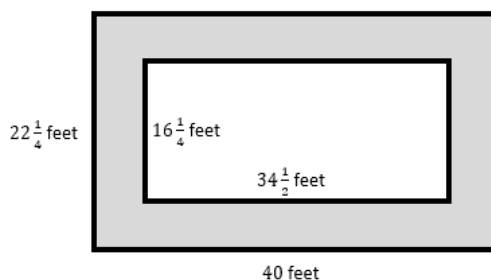
Instructional Item 1

A pentagon is shown. What is the area, in square inches, of the pentagon? Image not to scale.



Instructional Item 2

Mr. Moretti wants to cover the walkway around his swimming pool with tile. Determine how many square feet of tile he will need to cover the shaded portion of the diagram.



**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*